

Adaptive Neural Impedance Control of a Robotic Manipulator With Input Saturation

Wei He, *Member, IEEE*, Yiting Dong, *Student Member, IEEE*, and Changyin Sun

Abstract—In this paper, adaptive impedance control is developed for an n -link robotic manipulator with input saturation by employing neural networks. Both uncertainties and input saturation are considered in the tracking control design. In order to approximate the system uncertainties, we introduce a radial basis function neural network controller, and the input saturation is handled by designing an auxiliary system. By using Lyapunov's method, we design adaptive neural impedance controllers. Both state and output feedbacks are constructed. To verify the proposed control, extensive simulations are conducted.

Index Terms—Adaptive neural network (NN) control, impedance control, input saturation, learning control, nonlinear system, robot.

I. INTRODUCTION

IN RECENT years, the control design and stability analysis of a robot have received considerable attention. The position or force control systems cannot be formulated easily in order to control the robotic manipulator, given that the position and force control objectives cannot be considered separately [1]. An impedance control is proposed so that the control objective should not track the position or force trajectory only but rather involve the regulation of the impedance of the robot end effector which relates both position and force [2]–[4]. The main challenge of the impedance control design is the situation when it fails to provide adequate power due to the input nonlinearities, especially input saturation. Saturation nonlinearity can be caused by the constraints of magnitude and the rate of the actuator inputs [5]. Higher overshoot and larger tracking errors may result from neglecting the input saturation effect for a robotic system. It is therefore crucial to design an impedance controller which

can compensate for the input saturation effect of the robotic system and maintain the stability of the robot.

During the past decade, the control system of the robotic manipulator has been developed in a variety of robotic applications, such as rehabilitation, home service, education, and entertainment. Most of the robotic manipulator systems are integrated with fixed platforms, resulting in achieving physical robot–environment interaction. The main challenge in this field is to investigate the physical robot–environment interaction problem intensively, which has become popular on the development side [6], [7]. Focusing on the need of safe, fast, and accurate interaction, the position or velocity tracking method is not essential in the robotic control task. Impedance control was first proposed by Hogan [2]. The basic principle of this proposed method is to regulate the interaction between the manipulator end effector and the force exerted on the environment by specifying a desired relation or impedance force, rather than track the motion and force trajectory [8]. Many works have been done on learning impedance control in robotic platforms. In [9], an iterative learning impedance control problem is formulated and solved for robotic manipulators. The learning impedance control is designed such that the robotic control system has the ability to follow the desired response specified by the target model as the actions are repeated. A robust impedance control is proposed in [2], and this algorithm regulates the dynamic behavior at the interaction point. In [10], two adaptive impedance controllers are presented, and it assumes that some parameters in robotic manipulator dynamics may be uncertain. The robotic system is stable and the tracking performance is guaranteed in the above methods, but these algorithms all require knowledge of robotic dynamics.

For the purpose of solving the problem, learning control is presented which requires less knowledge [11]–[14]. In [15], adaptive neural networks (NNs) are used to approximate the unknown model of a rehabilitation robot. Colbaugh *et al.* [16] presented an adaptive scheme for controlling the end-effector impedance of the robot manipulator. Adaptive control has been developed using a variable structure method with the presence of uncertainties in [17]. In [18], target impedance reference trajectory is used to develop an NN impedance control efficiently. In [19], adaptive control strategies are presented for mobile manipulators in the presence of uncertainties and disturbances. For a class nonlinear systems, an adaptive fuzzy control scheme [20] and adaptive NN control [21] are proposed to realize the control objective. A hybrid motion/force control based on an adaptive NN is proposed successfully in [22] to compensate for the perturbations. In [23], a stable

Manuscript received February 5, 2015; accepted April 2, 2015. Date of publication June 1, 2015; date of current version February 12, 2016. This work was supported in part by the National Natural Science Foundation of China under Grant 61203057 and Grant 61125306, in part by the National Basic Research Program of China (973 Program) under Grant 2014CB744206, in part by the National High Technology Research and Development Program of China (863 Program) under Grant 2015AA042304, and in part by the Fundamental Research Funds for the China Central Universities of the University of Electronic Science and Technology of China under Grant ZYGX2013Z003 and Grant Y03001023601034. This paper was recommended by Associate Editor Z. Liu.

W. He and Y. Dong are with the School of Automation Engineering and Center for Robotics, University of Electronic Science and Technology of China, Chengdu 611731, China (e-mail: weihe@ieee.org).

C. Sun is with the School of Automation, Southeast University, Nanjing 210096, China.

Color versions of one or more of the figures in this paper are available online at <http://ieeexplore.ieee.org>.

Digital Object Identifier 10.1109/TSMC.2015.2429555

adaptive NN tracking control design technique is proposed for nonaffine nonlinear systems. In [24], an adaptive control for a class of nonlinear stochastic systems with unknown functions is studied. Lewis *et al.* [25] and Chen *et al.* [26] have initially applied backstepping control of nonlinear systems using NN to a robotic platform. NNs have an important property of function approximation and can be used to approximate nonlinear functions and unknown system dynamics [27]–[30]. In [31], to realize biped walking, a neural learning control system considering the energy cost is presented.

Those proposed controllers do not require the knowledge of the structure of robotic systems and are implemented without calculation of the robot inverse kinematic transformation. In most of the control algorithms, it is assumed that the system states are available. However, it may not be practical to measure all the values of system states. In [32], two adaptive NN decentralized output feedback control schemes are proposed for a class of nonlinear systems. Liu *et al.* [33] proposed an adaptive output feedback control for uncertain nonlinear single-input single-output systems. In [34], an adaptive neural output feedback tracking control of nonlinear multiple-input multiple-output systems is studied. In [35] and [36], the high gain observer is used to estimate unmeasurable system states effectively. A reduced-order observer is designed to estimate immeasurable states in [37].

Another challenge of the robotic manipulator control design is how to deal with the input saturation which may result in higher overshoot and larger tracking errors. A number of the methods are being used to handle the control design related with saturation. Santibañez *et al.* [38] proposed a saturated nonlinear PID controller for industrial robotic manipulators. However, the controller is based on the assumption that the information of saturator and system dynamics are known. Huang *et al.* [39] deal with the problem of global tracking and stabilization control of a robot, in the presence of input saturation and external disturbances. In [40], a learning control scheme is proposed for nonlinear uncertain systems with input saturation successfully. In [41] and [42], the adaptive control is presented to handle the external disturbance, saturation and deadzone for nonlinear systems. Liu *et al.* [43] investigated the unknown direction hysteresis model with adaptive NN control. Adaptive NN control is presented to compensate the effect of output constraint for an uncertain robot in [44]. In [45], adaptive tracking control is presented for a class of uncertain nonlinear systems with input saturation. In [46], adaptive fuzzy controllers are designed via backstepping for nonlinear strict feedback systems with unknown deadzone and immeasurable states. Guo *et al.* [47] proposed two memristor-based recurrent NNs with time delay via static or dynamic coupling. In [48], an adaptive NN control method for the nonlinear multiagent systems with time-delay is proposed. In [49], in the presence of input saturation and unknown external disturbance, two robust adaptive control algorithms are developed. Besides the works on ordinary differential equation systems with input saturation, partial differential equation systems with constraints [50], [51] have also been studied in recent years. He and Ge [52] presented boundary control to handle the effect of output constraint for the string system by Lyapunov method.

In order to suppress the riser's vibration in the presence of top tension constraint, a boundary controller is studied via integral Lyapunov function in [53]. In addition, some output information is not measurable in actual platforms. In this paper, an auxiliary system is introduced to handle the effect of saturation.

Based on the discussion above, we present adaptive neural impedance control for robot–environment interaction. The primary contributions of this paper are summarized as follows.

- 1) To solve the input saturation and external disturbance, we have designed an effective impedance controller which based on the Lyapunov function. Apart from that, the structures in both state and output feedbacks are presented.
- 2) A learning approach for robotic control systems is proposed, which does not require knowledge of robotic dynamics. Furthermore, the stability of closed-loop system can be ensured using Lyapunov stability theorem.
- 3) The bounds of NN approximation errors, NN weights and radial basis functions (RBFs) are not necessarily made known for control design in the design process.

The rest of this paper is organized as follows. In Section II, the preliminaries and system dynamics are presented. Section III illustrates the direct adaptive neural impedance control design process under the state and output feedback methods. Section IV shows the performance of proposed methods by simulations. The last section concludes this paper.

II. PRELIMINARIES AND PROBLEM FORMULATION

A. Preliminaries

Lemma 1 [54]: For bounded initial conditions, if there exists a C^1 continuous and positive definite Lyapunov function $V(x)$ satisfying $\kappa_1(\|x\|) \leq V(x) \leq \kappa_2(\|x\|)$, such that $\dot{V}(x) \leq -\kappa V(x) + c$, where $\kappa_1, \kappa_2 : \mathbb{R}^n \rightarrow \mathbb{R}$ are class $\mathcal{K}$ functions, κ and c are two positive constants, then the solution $x(t)$ is uniformly bounded.

Lemma 2 [55]: Consider the basis functions of Gaussian RBFNN with $\hat{Z}$ being the input vector, if $\hat{Z} = Z - \epsilon \tilde{\psi}$, where $\tilde{\psi}$ is a bounded vector and constant $\epsilon > 0$, then we have

$$s_j(\hat{Z}) = \exp \left[\frac{-\left(\hat{Z} - \mu_j\right)^T \left(\hat{Z} - \mu_j\right)}{\eta_j^2} \right], \quad j = 1, 2, \dots, l$$

$$S(\hat{Z}) = S(Z) + \epsilon S_t \quad (1)$$

where S_t is a bounded vector function.

Lemma 3 [56]–[58] (*Rayleigh–Ritz Theorem*): Let $A \in \mathbb{R}^{n \times n}$ be a real, symmetric, and positive-definite matrix; therefore, all the eigenvalues of A are real and positive. Let $\lambda_{\min}$ and $\lambda_{\max}$ denote the minimum and maximum eigenvalue of A , respectively; then for $\forall x \in \mathbb{R}^n$, we have

$$\lambda_{\min} \|x\|^2 \leq x^T A x \leq \lambda_{\max} \|x\|^2 \quad (2)$$

where $\|\cdot\|$ denotes the standard Euclidean norm.

In practice, the exact model of robot dynamics is not known. To approximate the exact values of dynamic terms, a Gaussian RBFNN can be utilized as it can approximate the smooth function [1].

B. Dynamics of the Robotic System

When the robot comes in contact with the environment, an interactive force develops between the robot and environment according to some user-defined dynamics, the target impedance. The target impedance between the robot position x and the vector of forces and moments exerted by the robot on the environment $\tau_e \in \mathbb{R}^n$ is expressed as [1]

$$D_d(\ddot{x}_c - \ddot{x}) + C_d(\dot{x}_c - \dot{x}) + G_d(x_c - x) = \tau_e \quad (3)$$

where $x_c \in \mathbb{R}^n$ are the commanded trajectories which are bounded, smooth and twice differentiable, and the D_d , C_d , and $G_d \in \mathbb{R}^{n \times n}$ are also specified by the user. D_d , C_d , and G_d are the mass, damping, and stiffness matrices of the mass-spring damper system which provides the impedance relationship between the robot contact forces τ_e and the position errors $x_c - x$.

It can be easily seen that when the robot is required to move in free space, we have $x_c = x$, $\forall t > 0$ and $\tau_e = 0$. On the other hand, while in contact with the environment, the end-effector response is to comply with the contact forces which defined by the target impedance (3). In general, the matrices D_d , C_d , and G_d are diagonal with constant positive elements. Nevertheless, there is considerable freedom to choose these matrices, which is one of the attractive features of the impedance control scheme.

The dynamics of an n -link rigid robotic system in Cartesian space is described by [1]

$$D(q)\ddot{x} + C(q, \dot{q})\dot{x} + G(q) = S(\tau) - \tau_e \quad (4)$$

where $q \in \mathbb{R}^n$ is the coordinate, $\tau \in \mathbb{R}^n$ is the applied joint torque, $D(q) \in \mathbb{R}^{n \times n}$ is a symmetric positive definite inertia matrix, $C(q, \dot{q})\dot{x} \in \mathbb{R}^n$ denotes the Centripetal and Coriolis torques, and $G(q) \in \mathbb{R}^n$ is the gravitational force, $\tau_e \in \mathbb{R}^n$ is the vector of constraint force exerted by the environment, which is 0 when there is no contact between the manipulator and environment.

Property 1 [1], [59]: The matrix $D(q)$ is symmetric and positive definite.

Property 2 [1], [59]: The matrix $\dot{D}(q) - 2C(q, \dot{q})$ is skew-symmetric.

The desired impedance model given in the joint space is given as

$$D_d(\ddot{x}_d - \ddot{x}) + C_d(q, \dot{q})(\dot{x}_d - \dot{x}) + G_d(x_d - x) = -\tau_e \quad (5)$$

where x_d is the desired trajectory, D_d , C_d , and G_d are the desired inertia, damping, and stiffness matrices, respectively.

Assumption 1: The desired trajectories $\|x_d\| \leq d$ with known constant $d > 0$ is smooth and differentiable. Therefore, define the time derivative of x_d as: $\|\dot{x}_d\| \leq d_1$, $\|\ddot{x}_d\| \leq d_2$.

The control objective is to design control torques such that the impedance of the whole system tracks the given target impedance model (5).

III. CONTROL DESIGN

Both state and output feedback schemes are presented in this section. The impedance control is aimed to reduce the saturation effect and track the desired trajectory.

A. Full State Feedback Control

We first consider the case where full state information x is available. Adaptive NN control is applied to approximate the unknown model of the robotic manipulator. Define the Cartesian space signals as

$$z_1 = x_d - x \quad (6)$$

$$\alpha_1 = \dot{x}_d + K_1 z_1 \quad (7)$$

$$z_2 = \alpha_1 - \dot{x} = \dot{z}_1 + K_1 z_1. \quad (8)$$

Taking the contact force τ_e into consideration, the Cartesian space dynamic NN based on adaptive control is modified, supposing the dynamic model and the desired impedance model are completely known, by adding the contact force τ_e , to

$$\tau(t) = -z_1 + \hat{D}\dot{\alpha}_1 + \hat{C}\alpha_1 + \hat{G} + K_p(z_2 + \zeta) + K_r \text{sgn}(z_2) + \tau_e \quad (9)$$

where $K_r = \text{diag}[k_{r_{ii}}]$, $\hat{D} = D$, $\hat{C} = C$, $\hat{G} = G$, and $\Delta\tau = S(\tau) - \tau(t)$. In actual implementation, the control input τ that must be applied to the joint.

Substituting (9) into (4), we have

$$-D\dot{z}_2 = -z_1 + Cz_2 + (\hat{D} - D)\dot{\alpha}_1 + (\hat{C} - C)\alpha_1 + \Delta\tau + (\hat{G} - G) + K_p(z_2 + \zeta) + K_r \text{sgn}(z_2) \quad (10)$$

where $\Delta\tau = S(\tau) - \tau(t)$. The following auxiliary design system is used to reduce the saturation effects:

$$\dot{\zeta} = \begin{cases} -K_\zeta \zeta - \frac{|z_2^T \Delta\tau| + \frac{1}{2} \Delta\tau^T \Delta\tau}{\|\zeta\|^2} \zeta + \Delta\tau, & \|\zeta\| \geq \mu \\ 0, & \|\zeta\| < \mu \end{cases} \quad (11)$$

where $K_\zeta = K_\zeta^T > 0$, μ is a small positive value and $\zeta \in \mathbb{R}^n$ is the state of auxiliary design system.

Considering Lyapunov function candidate $V_1 = (1/2)z_1^T z_1 + (1/2)z_2^T D z_2 + (1/2)\zeta^T \zeta$ and the above auxiliary design function (11). Time derivative of V_1 is

$$\dot{V}_1 = z_1^T \dot{z}_1 + z_2^T D \dot{z}_2 + \frac{1}{2} z_2^T \dot{D} z_2 + \zeta^T \dot{\zeta}. \quad (12)$$

Since $\zeta^T \Delta\tau \leq 1/2 \zeta^T \zeta + 1/2 \Delta\tau^T \Delta\tau$, we have $\zeta^T \dot{\zeta} = -\zeta^T K_\zeta \zeta - |z_2^T \Delta\tau| - 0.5 \Delta\tau^T \Delta\tau + \zeta^T \Delta\tau \leq -\zeta^T K_\zeta \zeta - |z_2^T \Delta\tau| + 1/2 \zeta^T \zeta$. Time derivative of V_1 will then be

$$\dot{V}_1 \leq -z_1^T K_1 z_1 - \frac{1}{2} z_2^T (2K_p - I) z_2 - \frac{1}{2} \zeta^T (2K_\zeta - I - K_p^T K_p) \zeta. \quad (13)$$

To ensure the closed loop stability, control parameters K_p and K_ζ should fulfill the following criteria: $K_1 > 0$, $2K_p - I > 0$, and $2K_\zeta - I - K_p^T K_p > 0$. So $\dot{V}_1$ will be negative definite.

Since the uncertainties exist in $D(q)$, $C(q, \dot{q})$, and $G(q)$, the model-based control design may not be realizable. Thus, the above model-based control is not applicable for the robot with uncertainties. To overcome the challenge, NNs are utilized to approximate the uncertainties and improve the performance of the system via the online estimation.

Impedance control based on NN can improve the performance of the system even though the impedance model of

the system is time-varying. We will design the adaptive NN impedance control to approximate the unknown model of the constrained robot and adapt interactions between robots and humans.

We design the controller as

$$\tau(t) = -z_1 + \hat{W}_D^T S_D(Z_D) \dot{\alpha}_1 + \hat{W}_C^T S_C(Z_C) \alpha_1 + \tau_e + \hat{W}_G^T S_G(Z_G) + K_p(z_2 + \zeta) + K_r \text{sgn}(z_2). \quad (14)$$

The NN updating laws are designed as

$$\dot{\hat{W}}_{Dk} = \Gamma_{Dk} (S_{Dk} \dot{\alpha}_1 z_{2k} - \sigma_{Dk} \hat{W}_{Dk}) \quad (15)$$

$$\dot{\hat{W}}_{Ck} = \Gamma_{Ck} (S_{Ck} \alpha_1 z_{2k} - \sigma_{Ck} \hat{W}_{Ck}) \quad (16)$$

$$\dot{\hat{W}}_{Gk} = \Gamma_{Gk} (S_{Gk} z_{2k} - \sigma_{Gk} \hat{W}_{Gk}) \quad (17)$$

where $\Gamma_{Dk} > 0$, $\Gamma_{Ck} > 0$, $\Gamma_{Gk} > 0$, and σ_i is a small positive constant for improving the robustness [51]. $\hat{W}_D^T S_D(Z_D)$ is an approximation of $W_D^{*T} S_D(Z_D)$, $\hat{W}_C^T S_C(Z_C)$ is an approximation of $W_C^{*T} S_C(Z_C)$ and $\hat{W}_G^T S_G(Z_G)$ is an approximation of $W_G^{*T} S_G(Z_G)$ and

$$W_D^{*T} S_D(Z_D) = D + \epsilon_D \quad (18)$$

$$W_C^{*T} S_C(Z_C) = C + \epsilon_C \quad (19)$$

$$W_G^{*T} S_G(Z_G) = G + \epsilon_G. \quad (20)$$

Substituting the design controller (14) into (4), we have

$$-D\dot{z}_2 = \tilde{W}_D^T S_D \dot{\alpha}_1 + \tilde{W}_C^T S_C \alpha_1 + \tilde{W}_G^T S_G + K_p(z_2 + \zeta) + K_r \text{sgn}(z_2) + E_x + \Delta\tau - z_1 + Cz_2 \quad (21)$$

where $K_r = \text{diag}[k_{rii}]$ and $E_x = \epsilon_D \dot{\alpha}_1 + \epsilon_C \alpha_1 + \epsilon_G$. Considering the Lyapunov function candidate

$$V_2 = \frac{1}{2} z_1^T z_1 + \frac{1}{2} z_2^T D z_2 + \frac{1}{2} \zeta^T \zeta + \frac{1}{2} \sum_{k=0}^{\infty} \tilde{W}_{Dk}^T \Gamma_{Dk}^{-1} \tilde{W}_{Dk} + \frac{1}{2} \sum_{k=0}^{\infty} \tilde{W}_{Ck}^T \Gamma_{Ck}^{-1} \tilde{W}_{Ck} + \frac{1}{2} \sum_{k=0}^{\infty} \tilde{W}_{Gk}^T \Gamma_{Gk}^{-1} \tilde{W}_{Gk}. \quad (22)$$

Time derivative of V_2 is

$$\begin{aligned} \dot{V}_2 = & -z_1^T K_1 z_1 - z_2^T C z_2 \\ & - z_2^T (\tilde{W}_D^T S_D \dot{\alpha}_1 + \tilde{W}_C^T S_C \alpha_1 \\ & + \tilde{W}_G^T S_G + K_p(z_2 + \zeta) + K_r \text{sgn}(z_2) + \Delta\tau + E_x) \\ & + \frac{1}{2} z_2^T \dot{D} z_2 + \zeta^T \dot{\zeta} + \sum_{k=0}^{\infty} \tilde{W}_{Dk}^T S_{Dk} \dot{\alpha}_1 z_{2k} \\ & + \sum_{k=0}^{\infty} \tilde{W}_{Ck}^T S_{Ck} \alpha_1 z_{2k} + \sum_{k=0}^{\infty} \tilde{W}_{Gk}^T S_{Gk} z_{2k} \\ & - \sum_{k=0}^{\infty} \sigma_{Dk} \tilde{W}_{Dk}^T \hat{W}_{Dk} - \sum_{k=0}^{\infty} \sigma_{Ck} \tilde{W}_{Ck}^T \hat{W}_{Ck} \\ & - \sum_{k=0}^{\infty} \sigma_{Gk} \tilde{W}_{Gk}^T \hat{W}_{Gk} \end{aligned} \quad (23)$$

where Property 2 has been used, and noting that

$$z_2^T \tilde{W}_D^T S_D \dot{\alpha}_1 = \sum_{k=0}^{\infty} \tilde{W}_{Dk}^T S_{Dk} \dot{\alpha}_1 z_{2k} \quad (24)$$

$$z_2^T \tilde{W}_C^T S_C \alpha_1 = \sum_{k=0}^{\infty} \tilde{W}_{Ck}^T S_{Ck} \alpha_1 z_{2k} \quad (25)$$

$$z_2^T \tilde{W}_G^T S_G = \sum_{k=0}^{\infty} \tilde{W}_{Gk}^T S_{Gk} z_{2k}. \quad (26)$$

Since $-\tilde{W}^T \hat{W} = -\tilde{W}^T (W^* + \tilde{W}) = -\tilde{W}^T \tilde{W} - \tilde{W}^T W^*$ and $-\tilde{W}^T W^* \leq (1/2)(\tilde{W}^T \tilde{W} + W^{*T} W^*)$, we have $-\tilde{W}^T \hat{W} \leq -(1/2)\tilde{W}^T \tilde{W} + (1/2)W^{*T} W^*$. Meanwhile, considering (24)–(26), with $k_{rii} \geq \|E_{xi}\|$, we have

$$\begin{aligned} \dot{V}_2 \leq & -z_1^T K_1 z_1 - \frac{1}{2} z_2^T (2K_p - I) z_2 \\ & - \frac{1}{2} \zeta^T (2K_\zeta - I - K_p^T K_p) \zeta \\ & - \frac{1}{2} \left(\sum_{k=0}^{\infty} \sigma_{Dk} \tilde{W}_{Dk}^T \tilde{W}_{Dk} + \sum_{k=0}^{\infty} \sigma_{Ck} \tilde{W}_{Ck}^T \tilde{W}_{Ck} \right. \\ & \quad \left. + \sum_{k=0}^{\infty} \sigma_{Gk} \tilde{W}_{Gk}^T \tilde{W}_{Gk} \right) \\ & + \frac{1}{2} \left(\sum_{k=0}^{\infty} \sigma_{Dk} W_{Dk}^{*T} W_{Dk}^* + \sum_{k=0}^{\infty} \sigma_{Ck} W_{Ck}^{*T} W_{Ck}^* \right. \\ & \quad \left. + \sum_{k=0}^{\infty} \sigma_{Gk} W_{Gk}^{*T} W_{Gk}^* \right) \\ \leq & -\kappa V_2 + C \end{aligned} \quad (27)$$

where

$$\begin{aligned} \kappa = & \min \left(2K_1, \lambda_{\min}(2K_p - I), \lambda_{\min}(2K_\zeta - I - K_p^T K_p) \right. \\ & \times \left. \frac{\sigma_{Dk}}{\lambda_{\max}(\Gamma_{Dk}^{-1})}, \frac{\sigma_{Ck}}{\lambda_{\max}(\Gamma_{Ck}^{-1})}, \frac{\sigma_{Gk}}{\lambda_{\max}(\Gamma_{Gk}^{-1})} \right) \end{aligned} \quad (28)$$

$$C = \frac{1}{2} \left(\sum_{k=0}^{\infty} \sigma_{Dk} W_{Dk}^{*T} W_{Dk}^* + \sum_{k=0}^{\infty} \sigma_{Ck} W_{Ck}^{*T} W_{Ck}^* + \sum_{k=0}^{\infty} \sigma_{Gk} W_{Gk}^{*T} W_{Gk}^* \right). \quad (29)$$

To ensure the closed loop stability, control parameters K_1 , K_p , and K_ζ should fulfill the following criteria: $K_1 > 0$, $2K_p - I > 0$ and $2K_\zeta - I - K_p^T K_p > 0$. So $\dot{V}_2$ will be negative definite. Therefore, we have the following theorem.

Remark 1 [1]: In the proposed controller (14), the gain of the term K_r , is chosen such that $k_{rii} > \|E_{xi}\|$. For stability, K_r is usually chosen to be conservatively large. This is not very desirable due to the chattering introduced. The control term K_r can be changed to

$$K_r = k_D \dot{\alpha}_1 + k_C \alpha_1 + k_G \quad (30)$$

where $k_D \geq \|\epsilon_D\|$, $k_C \geq \|\epsilon_C\|$, and $k_G \geq \|\epsilon_G\|$. The gain has the following advantages: 1) the values of elements k_D , k_C , and k_G are only needed to be large enough to suppress the bounded approximation NN errors ϵ_D , ϵ_C , and ϵ_G , respectively and 2) it is a function of $\dot{\alpha}_1$ and α_1 so that it decreased as $\dot{\alpha}_1$ and α_1 diminish.

Theorem 1: For system dynamics described by (4), under Assumption 1, and the control (14) with the adaptation laws (15)–(17), given that the initial conditions are bounded, we can conclude that the desired impedance is achieved and the impedance error will eventually converge to a small neighborhood around zero by appropriately choosing design parameters. Furthermore, tracking error z_1 converges asymptotically to the compact set $\Omega_{z_1} := \{z_1 \in \mathbb{R}^2 \mid \|z_1\| \leq \sqrt{B}\}$ where $B = 2(V_2^*(0) + C/\kappa)$ with κ and C given in (28) and (29).

The proof of Theorem 1 is given in the Appendix.

B. Output Feedback Control

The proposed control (9) is designed under the assumption that all outputs are measurable. However, some output information may not be measurable due to practical issues such as cost and dimensions. In this section, a high gain observer [60] is used to estimate the unmeasurable terms. According to [15], the unmeasurable state $\dot{x}$ can, therefore, be approximated as π_2/γ . The estimate of the unmeasurable state vector z_2 is designed as

$$\hat{z}_2 = \frac{\pi_2}{\varepsilon} - \alpha_1 \quad (31)$$

where the dynamics of π_2 are described as

$$\varepsilon \dot{\pi}_1 = \pi_2 \quad (32)$$

$$\varepsilon \dot{\pi}_2 = -\bar{\lambda}_1 \pi_2 - \pi_1 + x \quad (33)$$

where ε is any small constant. According to [15], there exist positive constants t^* and r_n such that $\forall t > t^*$, we have $|\beta_n| \leq \varepsilon r_n$. So the $\tilde{z}_2$ can be estimated as follows:

$$\tilde{z}_2 = \hat{z}_2 - z_2 = \frac{\pi_2}{\varepsilon} - \alpha_1 - \dot{x} + \alpha_1 = \beta_n. \quad (34)$$

The control design is rewritten as follows:

$$\begin{aligned} \tau(t) = & \hat{W}_D^T \hat{S}_D \dot{\alpha}_1 + \hat{W}_C^T \hat{S}_C \alpha_1 + \hat{W}_G^T \hat{S}_G \\ & + K_p(\hat{z}_2 + \zeta_2) + K_r \text{sgn}(\hat{z}_2) + \tau_e - z_1. \end{aligned} \quad (35)$$

The following auxiliary design system is used to examine the saturation effect:

$$\dot{\zeta}_2 = \begin{cases} -K_\zeta \zeta_2 - \frac{|\zeta_2^T \Delta \tau| + \frac{1}{2} \Delta \tau^T \Delta \tau}{\|\zeta_2\|^2} \zeta_2 + \Delta \tau, & \|\zeta_2\| \geq \mu \\ 0, & \|\zeta_2\| < \mu. \end{cases} \quad (36)$$

The NN updating laws are designed as

$$\dot{\hat{W}}_D = \Gamma_D (\hat{S}_D \dot{\alpha}_1 \hat{z}_2 - \sigma_D \hat{W}_D) \quad (37)$$

$$\dot{\hat{W}}_C = \Gamma_C (\hat{S}_C \alpha_1 \hat{z}_2 - \sigma_C \hat{W}_C) \quad (38)$$

$$\dot{\hat{W}}_G = \Gamma_G (\hat{S}_G \hat{z}_2 - \sigma_G \hat{W}_G). \quad (39)$$

Differentiating (8) yield $\dot{z}_2 = \dot{\alpha}_1 - \ddot{x} = \dot{\alpha}_1 - D^{-1}(S(\tau) - G - C\dot{x} - \tau_e)$. Considering the Lyapunov function candidate

$$\begin{aligned} V_3 = & \frac{1}{2} z_1^T z_1 + \frac{1}{2} z_2^T D z_2 + \frac{1}{2} \zeta_2^T \zeta_2 + \frac{1}{2} \tilde{W}_D^T \Gamma_D^{-1} \tilde{W}_D \\ & + \frac{1}{2} \tilde{W}_C^T \Gamma_C^{-1} \tilde{W}_C + \frac{1}{2} \tilde{W}_G^T \Gamma_G^{-1} \tilde{W}_G. \end{aligned} \quad (40)$$

Time derivative of V_3 is

$$\begin{aligned} \dot{V}_3 = & -z_1^T K_1 z_1 + z_2^T ((D - \hat{D})\dot{\alpha}_1 + (C - \hat{C})\alpha_1 \\ & + (G - \hat{G}) - (C + K_p)z_2) - z_2^T K_p \zeta_2 - z_2^T K_r \text{sgn}(z_2) \\ & - z_2^T \Delta \tau + \frac{1}{2} z_2^T \dot{D} z_2 + \zeta_2^T \dot{\zeta}_2 + \tilde{W}_D^T \Gamma_D^{-1} \dot{\tilde{W}}_D \\ & + \tilde{W}_C^T \Gamma_C^{-1} \dot{\tilde{W}}_C + \tilde{W}_G^T \Gamma_G^{-1} \dot{\tilde{W}}_G \end{aligned} \quad (41)$$

where

$$\begin{aligned} & z_2^T (D - \hat{D})\dot{\alpha}_1 + (C - \hat{C})\alpha_1 + (G - \hat{G}) \\ & = z_2^T (W_D^{*T} S_D - \tilde{W}_D^T \hat{S}_D - W_D^{*T} \hat{S}_D)\dot{\alpha}_1 \\ & \quad + z_2^T (W_C^{*T} S_C - \tilde{W}_C^T \hat{S}_C - W_C^{*T} \hat{S}_C)\alpha_1 \\ & \quad + z_2^T (W_G^{*T} S_G - \tilde{W}_G^T \hat{S}_G - W_G^{*T} \hat{S}_G). \end{aligned} \quad (42)$$

Considering the effect of estimation error $\tilde{z}_2$, $V = V_3 + V_{\text{obs}}$ and $V_{\text{obs}} = 1/2 \tilde{z}_2^T \tilde{z}_2$. Based on (33), we have

$$V_{\text{obs}} \leq \frac{1}{2} \gamma^2 h_2^2. \quad (43)$$

We just need to check the stability of V_3 term as V_{obs} is stable. According to [1] and [55], we have

$$\hat{S}_D - S_D = \bar{\gamma}_D S_D, \hat{S}_C - S_C = \bar{\gamma}_C S_C, \hat{S}_G - S_G = \bar{\gamma}_G S_G \quad (44)$$

where S_D , S_C , and S_G are bounded vector functions with satisfying $\|S_D\| < S_D^*$, $\|S_C\| < S_C^*$, $\|S_G\| < S_G^*$ and $\bar{\gamma}_D > 0$, $\bar{\gamma}_C > 0$ and $\bar{\gamma}_G > 0$.

Considering the first term of (42), we have

$$\begin{aligned} & z_2^T (W_D^{*T} S_D - \tilde{W}_D^T \hat{S}_D - W_D^{*T} \hat{S}_D)\dot{\alpha}_1 \\ & = (z_2^T \tilde{W}_D^T \hat{S}_D \ddot{x}_d - z_2^T \tilde{W}_D^T \hat{S}_D K_1^2 z_1 + z_2^T \tilde{W}_D^T \hat{S}_D K_1 z_2) \\ & \quad - z_2^T \tilde{W}_D^T \hat{S}_D \dot{\alpha}_1 - z_2^T W_D^{*T} \bar{\gamma}_D S_D \ddot{x}_d \\ & \quad + z_2^T W_D^{*T} \bar{\gamma}_D S_D K_1^2 z_1 - z_2^T W_D^{*T} \bar{\gamma}_D S_D K_1 z_2. \end{aligned} \quad (45)$$

Substituting $\|\hat{S}_i\|^2 \leq l_i$ into the above function, we have

$$\begin{aligned} & z_2^T \tilde{W}_D^T \hat{S}_D \ddot{x}_d \leq \frac{l_D \theta_1 \|\gamma h_2\|^2}{4} \|\tilde{W}_D\|^2 + \frac{d_2^2}{\theta_1} \\ & - z_2^T \tilde{W}_D^T \hat{S}_D K_1^2 z_1 \leq \frac{l_D \theta_2 K_1^4 \|\gamma h_2\|^2}{4} \|\tilde{W}_D\|^2 + \frac{1}{\theta_2} z_1^T z_1 \\ & z_2^T \tilde{W}_D^T \hat{S}_D K_1 z_2 \leq \frac{l_D \theta_3 K_1^2 \|\gamma h_2\|^2}{4} \|\tilde{W}_D\|^2 + \frac{1}{\theta_3} z_2^T z_2. \end{aligned} \quad (46)$$

Considering the above function, we have

$$\begin{aligned} & -z_2^T W_D^{*T} \bar{\gamma}_D S_D \ddot{x}_d + z_2^T W_D^{*T} \bar{\gamma}_D S_D K_1^2 z_1 - z_2^T W_D^{*T} \bar{\gamma}_D S_D K_1 z_2 \\ & \leq \frac{1}{2} z_2^T z_2 + \frac{d_2^2}{2} \|W_D^*\|^2 \|\bar{\gamma}_D S_D\|^2 + \frac{1}{2} z_2^T z_2 \\ & \quad + \frac{1}{2} z_1^T \|W_D^*\|^2 \|\bar{\gamma}_D S_D\|^2 \|K_1\|^4 z_1 \\ & \quad + \frac{1}{2} z_2^T z_2 + \frac{1}{2} z_2^T \|W_D^*\|^2 \|\bar{\gamma}_D S_D\|^2 \|K_1\|^2 z_2. \end{aligned} \quad (47)$$

Therefore, we have

$$\begin{aligned} & z_2^T (W_D^{*T} S_D - \tilde{W}_D^T \hat{S}_D - W_D^{*T} \hat{S}_D) \dot{\alpha}_1 \\ & \leq z_1^T \left(\frac{1}{\theta_2} + \frac{1}{2} \|W_D^*\|^2 \|\tilde{\gamma}_D S_D\|^2 \|K_1\|^4 \right) z_1 \\ & + z_2^T \left(\frac{1}{\theta_3} + \frac{3}{2} I + \frac{1}{2} \|W_D^*\|^2 \|\tilde{\gamma}_D S_D\|^2 \|K_1\|^2 \right) z_2 \\ & + \left(\frac{l_D \|\gamma h_2\|^2 (\theta_1 + \theta_2 K_1^4 + \theta_3 K_1^2)}{4} \right) \|\tilde{W}_D\|^2 \\ & + \left(\frac{d_2^2}{\theta_1} + \frac{d_2^2}{2} \|\tilde{\gamma}_D S_D\|^2 \|W_D^*\|^2 \right). \end{aligned} \quad (48)$$

Considering the second term of (42), we have

$$\begin{aligned} & z_2^T (W_C^{*T} S_C - \tilde{W}_C^T \hat{S}_C - W_C^{*T} \hat{S}_C) \alpha_1 \\ & \leq \frac{\theta_4}{4} l_C \|\gamma h_2\|^2 \|\tilde{W}_C\|^2 + \frac{d_1^2}{\theta_4} + \frac{1}{\theta_5} z_1^T z_1 \\ & + \frac{\theta_5}{4} l_C \|\gamma h_2\|^2 \|K_1\|^2 \|\tilde{W}_C\|^2 - \hat{z}_2^T \tilde{W}_C^T \hat{S}_C \alpha_1 \\ & + \frac{1}{2} z_2^T z_2 + \frac{d_1^2}{2} \|W_C^*\|^2 \|\tilde{\gamma}_C S_C\|^2 + \frac{1}{2} z_2^T z_2 \\ & + \frac{1}{2} z_1^T \|W_C^*\|^2 \|\tilde{\gamma}_C S_C\|^2 \|K_1\|^2 z_1. \end{aligned} \quad (49)$$

Considering the last term of (42), we have

$$\begin{aligned} & z_2^T (-\tilde{W}_G^T \hat{S}_G - W_G^{*T} \tilde{\gamma}_G S_G) \leq \frac{\theta_6}{4} l_G \|\gamma h_2\|^2 \|\tilde{W}_G\|^2 + \frac{1}{\theta_6} \\ & + \frac{1}{2} z_2^T z_2 + \frac{1}{2} \|W_G^*\|^2 \|\tilde{\gamma}_G S_G\|^2 \\ & - \hat{z}_2^T \tilde{W}_G^T \hat{S}_G. \end{aligned} \quad (50)$$

Therefore, time derivative of V_3 is

$$\begin{aligned} \dot{V}_3 & \leq z_1^T \left[\frac{\theta_2 + \theta_5}{\theta_2 \theta_5} + \frac{1}{2} (\|W_D^*\|^2 \|\tilde{\gamma}_D S_D\|^2 K_1^4 \right. \\ & \quad \left. + \|W_C^*\|^2 \|\tilde{\gamma}_C S_C\|^2 K_1^2) - K_1 \right] z_1 \\ & + z_2^T \left[\frac{1}{\theta_3} + 3.5I - K_p + \frac{1}{2} \|W_D^*\|^2 \|\tilde{\gamma}_D S_D\|^2 K_1^2 \right] z_2 \\ & + \frac{1}{2} \zeta_2^T (K_p^T K_p - 2K_\zeta + I) \zeta_2 + \left[\frac{l_G \theta_6 \|\gamma h_2\|^2}{4} \right] \|\tilde{W}_G\|^2 \\ & + \left[\frac{l_D \|\gamma h_2\|^2 (\theta_2 K_1^4 + \theta_3 K_1^2 + \theta_4)}{4} \right] \|\tilde{W}_D\|^2 \\ & + \left[\frac{l_C \|\gamma h_2\|^2 (\theta_4 + \theta_5 K_1^2)}{4} \right] \|\tilde{W}_C\|^2 \\ & + \frac{d_2^2}{2} \|\tilde{\gamma}_D S_D\|^2 \|W_D^*\|^2 + \frac{d_1^2}{2} \|\tilde{\gamma}_C S_C\|^2 \|W_C^*\|^2 \\ & + \frac{1}{2} \|\tilde{\gamma}_G S_G\|^2 \|W_G^*\|^2 + \frac{d_2^2}{\theta_1} + \frac{d_1^2}{\theta_4} + \frac{1}{\theta_6} \\ & - \sigma_D \tilde{W}_D^T \hat{W}_D - \sigma_C \tilde{W}_C^T \hat{W}_C - \sigma_G \tilde{W}_G^T \hat{W}_G. \end{aligned} \quad (51)$$

Since $-\tilde{W}_i^T \hat{W}_i = -\tilde{W}_i^T (W_i^* + \tilde{W}_i)$ and $-\tilde{W}_i^T W_i^* \leq (1/2)(\tilde{W}_i^T \tilde{W}_i + W_i^{*T} W_i^*)$, we have $-\tilde{W}_i^T \hat{W}_i \leq -(1/2)\tilde{W}_i^T \tilde{W}_i +$

$(1/2)W_i^{*T} W_i^*$. Substituting these inequalities into (51), we have

$$\begin{aligned} \dot{V}_3 & \leq z_1^T \left[\frac{\theta_2 + \theta_5}{\theta_2 \theta_5} + \frac{1}{2} (\|W_D^*\|^2 \|\tilde{\gamma}_D S_D\|^2 K_1^4 \right. \\ & \quad \left. + \|W_C^*\|^2 \|\tilde{\gamma}_C S_C\|^2 K_1^2) - K_1 \right] z_1 \\ & + \frac{1}{2} \zeta_2^T K_p^T K_p \zeta_2 \\ & + z_2^T \left[\frac{1}{\theta_3} + 3.5I - K_p + \frac{1}{2} \|W_D^*\|^2 \|\tilde{\gamma}_D S_D\|^2 K_1^2 \right] z_2 \\ & + \left[\frac{l_G \theta_6 \|\gamma h_2\|^2}{4} - \frac{\sigma_G}{2} \right] \|\tilde{W}_G\|^2 + \frac{d_2^2}{\theta_1} + \frac{d_1^2}{\theta_4} + \frac{1}{\theta_6} \\ & + \left[\frac{l_D \|\gamma h_2\|^2 (\theta_2 K_1^4 + \theta_3 K_1^2 + \theta_4)}{4} - \frac{\sigma_D}{2} \right] \|\tilde{W}_D\|^2 \\ & + \left[\frac{l_C \|\gamma h_2\|^2 (\theta_4 + \theta_5 K_1^2)}{4} - \frac{\sigma_C}{2} \right] \|\tilde{W}_C\|^2 + \frac{1}{2} \zeta_2^T \zeta_2 \\ & + \frac{1}{2} (d_2^2 \|\tilde{\gamma}_D S_D\|^2 + \sigma_D) \|W_D^*\|^2 - \zeta_2^T K_\zeta \zeta_2 \\ & + \frac{1}{2} (d_1^2 \|\tilde{\gamma}_C S_C\|^2 + \sigma_C) \|W_C^*\|^2 \\ & + \frac{1}{2} (\|\tilde{\gamma}_G S_G\|^2 + \sigma_G) \|W_G^*\|^2 \\ & \leq -\kappa V_3 + C \end{aligned} \quad (52)$$

where

$$\begin{aligned} \kappa & = \min \left(2\lambda_{\min} \left(-\frac{\theta_2 + \theta_5}{\theta_2 \theta_5} - \frac{1}{2} (\|W_D^*\|^2 \|\tilde{\gamma}_D S_D\|^2 K_1^4 \right. \right. \\ & \quad \left. \left. - \|W_C^*\|^2 \|\tilde{\gamma}_C S_C\|^2 K_1^2) + K_1 \right), \right. \\ & \quad 2\lambda_{\min} \left(-\frac{1}{\theta_3} - 3.5I + K_p - \frac{1}{2} \|W_D^*\|^2 \|\tilde{\gamma}_D S_D\|^2 K_1^2 \right), \\ & \quad \lambda_{\min} (-K_p^T K_p + 2K_\zeta - I), \\ & \quad \frac{l_D \|\gamma h_2\|^2 (\theta_2 K_1^4 + \theta_3 K_1^2 + \theta_4) - 2\sigma_D}{4\lambda_{\max}(\Gamma_{Dk}^{-1})}, \\ & \quad \frac{l_C \|\gamma h_2\|^2 (\theta_4 + \theta_5 K_1^2) - 2\sigma_C}{4\lambda_{\max}(\Gamma_{Ck}^{-1})}, \\ & \quad \left. \frac{l_G \theta_6 \|\gamma h_2\|^2 - 2\sigma_G}{4\lambda_{\max}(\Gamma_{Gk}^{-1})} \right) \end{aligned} \quad (53)$$

$$\begin{aligned} C & = \frac{1}{2} (\|\tilde{\gamma}_D S_D\|^2 d_2^2 + \sigma_D) \|W_D^*\|^2 + \frac{d_2^2}{\theta_1} \\ & + \frac{1}{2} (\|\tilde{\gamma}_C S_C\|^2 d_1^2 + \sigma_C) \|W_C^*\|^2 + \frac{d_1^2}{\theta_4} \\ & + \frac{1}{2} (\|\tilde{\gamma}_G S_G\|^2 + \sigma_G) \|W_G^*\|^2 + \frac{1}{\theta_6}. \end{aligned} \quad (54)$$

To ensure the closed loop stability, control parameters κ and C should fulfill the following criteria: $\kappa > 0$, $C > 0$. So $\dot{V}_3$ will be negative definite.

Theorem 2: For the system dynamics described by (4), under Assumption 1, and the control (35) with the adaptation laws (37)–(39), given that the initial conditions are bounded, we can conclude that the desired impedance is achieved and the impedance error will eventually converge to a small neighborhood around zero by appropriately choosing design parameters. Furthermore, tracking error z_1 converges asymptotically to the compact set $\Omega_{z_1} := \{z_1 \in \mathbb{R}^2 \mid \|z_1\| \leq \sqrt{B}\}$ where $B = 2(V_3^*(0) + C/\kappa)$ with κ and C given in (53) and (54).

The proof of Theorem 2 is straightforward using the method of proof in Theorem 1, and hence is omitted.

IV. SIMULATION STUDY

In this section, we will verify the effectiveness of the proposed adaptive neural control (14) and (35) for the 2-DOF robotic manipulator with input saturation. For a robotic system defined by (4) with two rotary degree of freedom, it can be expressed as

$$\begin{aligned} \dot{x}_1 &= x_2 \\ \dot{x}_2 &= D^{-1}[S(\tau) - \tau_e - G(x_1) - C(x_1, x_2)x_2] \\ y &= x_1 \end{aligned} \quad (55)$$

where as $x_1 = [q_1, q_2]^T$ and $x_2 = [\dot{q}_1, \dot{q}_2]^T$. Considering the 2-DOF robotic manipulator in a vertical plane, simulations are conducted to verify the effectiveness of the proposed control. Let m_i and l_i be the mass and length of link i , l_{ci} be the distance from joint $i - 1$ to the center of mass of link i , as indicated in the figure, and I_i be the moment of inertia of link i about an axis coming out of the page passing through the center of mass of link i .

The inertia matrix $D(x_1)$, centripetal and Coriolis torques $C(x_1, x_2)$, and gravitational force $G(x_1)$ are defined as

$$D(x_1) = \begin{bmatrix} p_1 + p_2 + 2p_3 \cos q_2 & p_2 + p_3 \cos q_2 \\ p_2 + p_3 \cos q_2 & p_2 \end{bmatrix} \quad (56)$$

$$C(x_1, x_2) = \begin{bmatrix} -p_3 \dot{q}_2 \sin q_2 & -p_3 (\dot{q}_1 + \dot{q}_2) \sin q_2 \\ p_3 \dot{q}_1 \sin q_2 & 0 \end{bmatrix} \quad (57)$$

$$G(x_1) = \begin{bmatrix} p_4 g \cos q_1 + p_5 g \cos(q_1 + q_2) \\ p_5 g \cos(q_1 + q_2) \end{bmatrix} \quad (58)$$

where

$$p_1 = m_1 l_{c1}^2 + m_2 l_1^2 + I_1 \quad (59)$$

$$p_2 = m_2 l_{c2}^2 + I_2 \quad (60)$$

$$p_3 = m_2 l_1 l_{c2} \quad (61)$$

$$p_4 = m_1 l_{c2} + m_2 l_1 \quad (62)$$

$$p_5 = m_2 l_{c2}. \quad (63)$$

Parameters of the robotic system are defined in Table I.

The commanded trajectory chosen for simulation is as shown in Fig. 1. We require the end effector to move along the solid part of the circle x_d in free space, and to move slide along the wall when in contact with the wall. The commanded

TABLE I
PARAMETERS OF THE ROBOT

Parameter	Description	Value
m_1	Mass of link 1	2.0 kg
m_2	Mass of link 2	0.85 kg
l_1	Length of link 1	0.35 m
l_2	Length of link 2	0.31 m
I_1	Inertia of link 1	$61.25 \times 10^{-3} \text{ kgm}^2$
I_2	Inertia of link 2	$20.42 \times 10^{-3} \text{ kgm}^2$

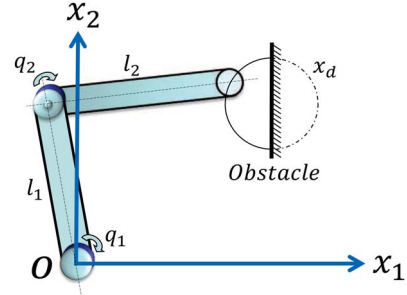


Fig. 1. Mechanical model of the robotic manipulator.

trajectory is described by

$$x_d(t) = \begin{bmatrix} x_{d1} \\ x_{d2} \end{bmatrix} = \begin{bmatrix} 0.2 - 0.2 \cos(\pi t) \\ 0.2 + 0.2 \sin(\pi t) \end{bmatrix}. \quad (64)$$

The radius of circle is 0.2 and the center is located at $x = [0.2 \ 0.2]^T$. The obstacle is located as $x_1 = 0.2$. The initial condition of the robot is

$$q(0) = [0.25 \ -0.05]^T \quad \dot{q}(0) = [0.0 \ 0.0]^T. \quad (65)$$

The unknown saturations are defined as $[-17.0 \ 17.0]$ and $[-15.0 \ 15.0]$ for q_1 and q_2 , respectively. The parameters of the target impedance are chosen as

$$D_d = \text{diag}[1.0] \quad C_d = \text{diag}[10.0] \quad G_d = \text{diag}[60.0]. \quad (66)$$

The gains of the control law are chosen as

$$K_p = \text{diag}[10.0] \quad K_1 = \text{diag}[6.0]. \quad (67)$$

The simulation objective is to examine whether the control scheme we designed for the system (4) is able to follow the desired trajectory (64). We have considered three cases for the simulation.

A. Case 1: PD Control

The PD control law is designed as: $\tau(t) = K_p(z_2 + \zeta) + K_r \text{sgn}(z_2)$. When the adaptive rules given by (15)–(17) are not activated, the control performance is studied at first. It is easy to see that the designed control action is effectively a simple PD control. Fig. 2 shows the position tracking of the robot and Fig. 3 gives the tracking errors. The control torques are given in Fig. 4. It can be observed from these results that, the PD control has significant tracking errors, especially the error of x_1 .

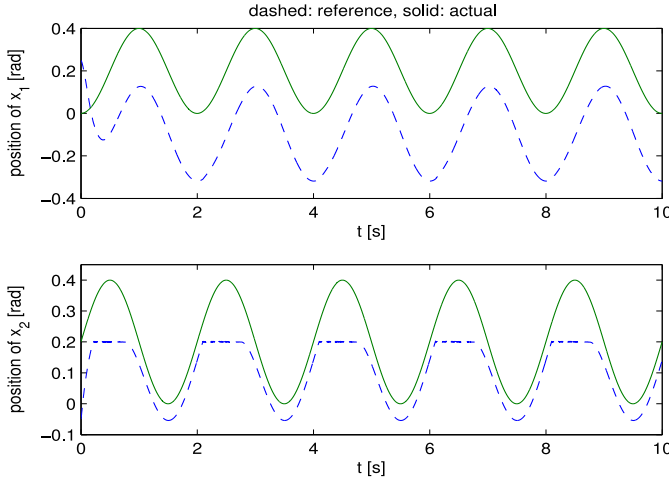


Fig. 2. Position tracking with PD control.

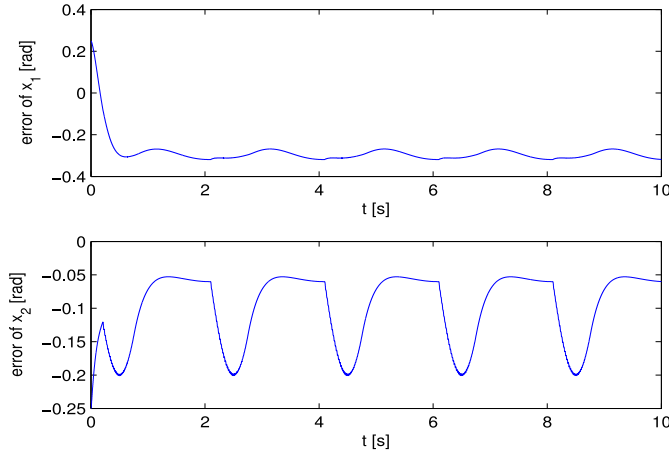


Fig. 3. Tracking errors with PD control.

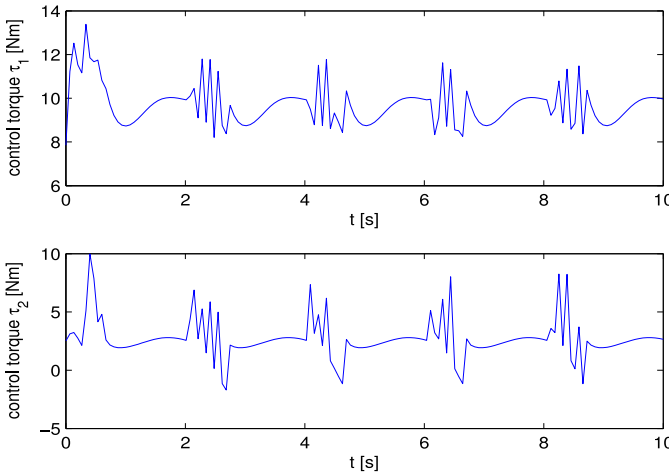


Fig. 4. Control signals with PD control.

B. Case 2: Adaptive Neural Impedance Control With State Feedback (14)

The new design is added to the simulation, which is involved in the state feedback control. For the situation when we do

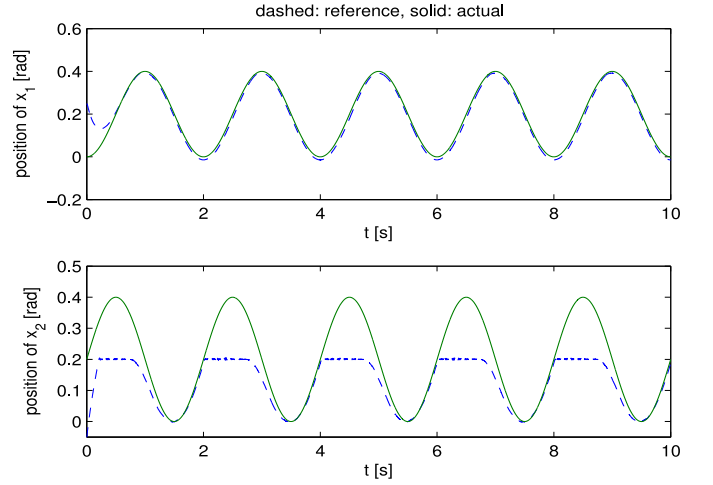


Fig. 5. Position tracking with state feedback control.

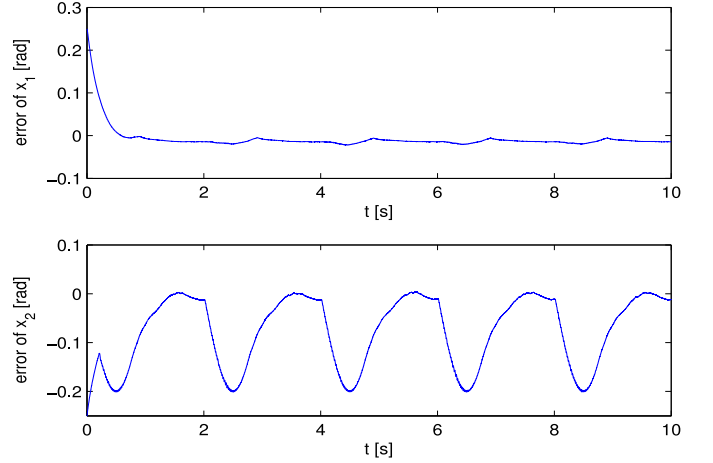


Fig. 6. Tracking errors with state feedback control.

not have complete information of the robotic system, the proposed control (14) can be implemented to approximate the system. For the approximation-based control, a number of 256 nodes are used for each $S_i(Z)$ with centers chosen in the area of $[-1, 1] \times [-1, 1] \times [-1, 1] \times [-1, 1] \times [-1, 1] \times [-1, 1] \times [-1, 1] \times [-1, 1]$, and the value of the center is set as -1 or 1 . The RBFNNs with $Z_D = [x_1, x_2, \alpha_1, \dot{\alpha}_1]$, $Z_C = [x_1, x_2, \alpha_1, \dot{\alpha}_1]$, and $Z_G = [x_1, x_2, \alpha_1, \dot{\alpha}_1]$, the number of nodes l_D , l_C , and l_G for S_D , S_C , and S_G is 2^8 . The gain matrix Γ is defined as $\Gamma_D = 100I_{256 \times 256}$, $\Gamma_C = 100I_{256 \times 256}$, and $\Gamma_G = 100I_{256 \times 256}$. $\sigma_D = 0.01$, $\sigma_C = 0.01$, and $\sigma_G = 0.01$ and the variance for all NNs is set to be 25. RBFNN centers for all NNs are evenly distributed in the domain of $[-1, 1] \times [-1, 1] \times [-1, 1] \times [-1, 1] \times [-1, 1] \times [-1, 1] \times [-1, 1] \times [-1, 1]$. Initial weights of the NNs are all zero. The same control gains as that in case 1 are used for the adaptive impedance control.

The position tracking and the errors of the manipulator are plotted in Figs. 5 and 6, respectively, and the control signals are given in Fig. 7. It is obvious that the tracking errors are much smaller than the nonadaptive case, thus, demonstrating the effectiveness of the adaptive NN impedance control in handling the unknown dynamic system with saturation effect.

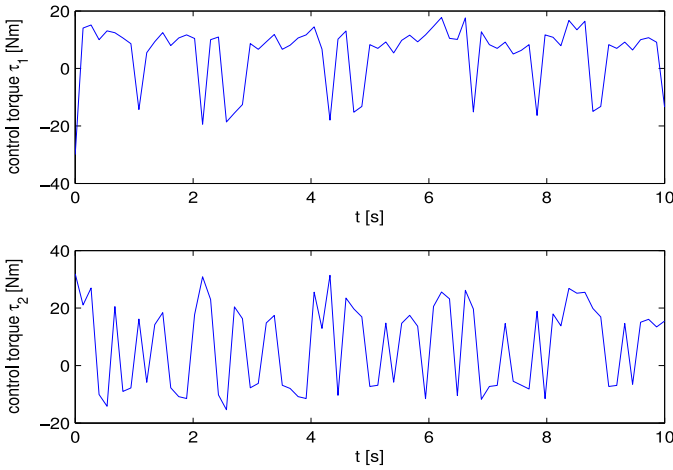


Fig. 7. Control signals with state feedback control.

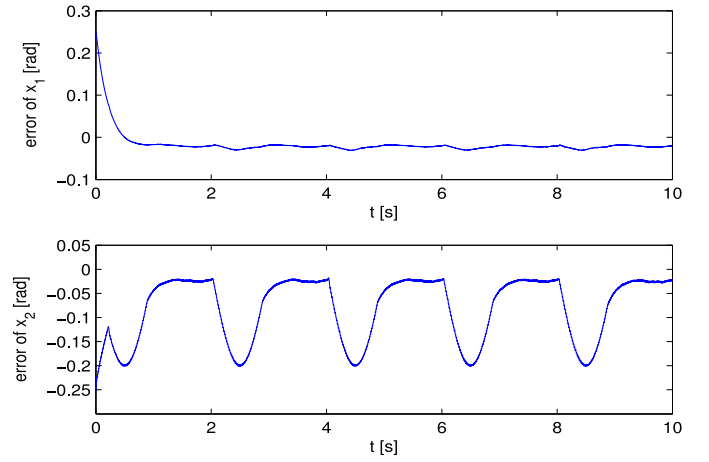


Fig. 9. Tracking errors with output feedback control.

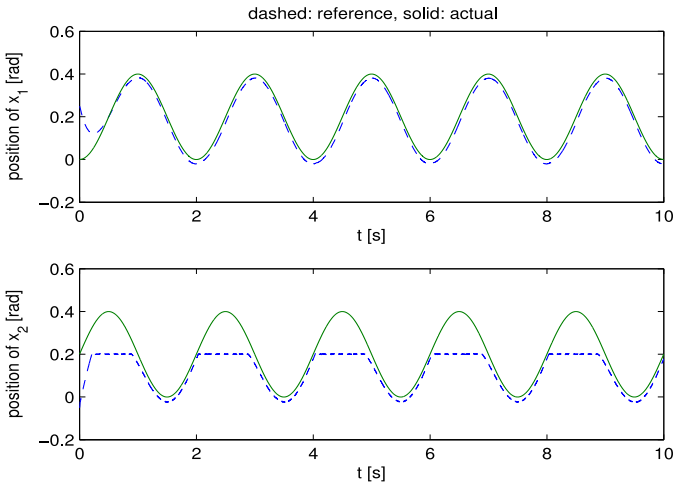


Fig. 8. Position tracking with output feedback control.

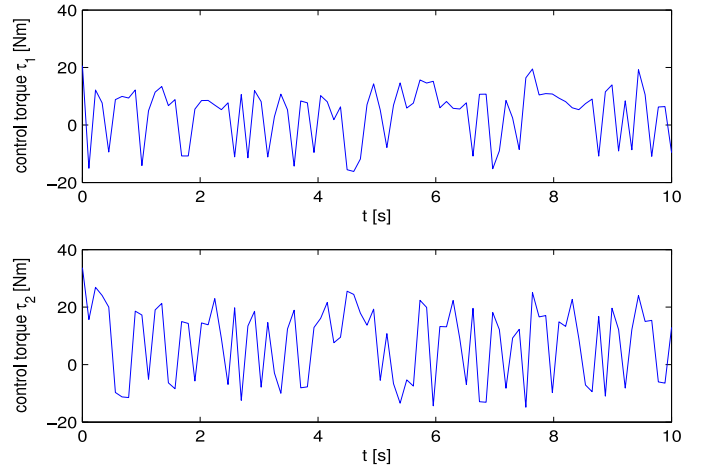


Fig. 10. Control signals with output feedback control.

C. Case 3: Adaptive Neural Impedance Control With Output Feedback (35)

With all parameters being the same as previously defined in case 2, the new parameters are added to the simulation: $\epsilon = 0.001$, $\lambda_1 = [3; 3]$, and initial terms $\pi_1 = 0$, $\pi_2 = 0$, $\dot{\pi}_1 = 0$, and $\dot{\pi}_2 = 0$.

Figs. 8 and 9 show the position tracking and the tracking errors of the robotic manipulator, respectively. Fig. 10 gives the control signals. There exists the tracking errors of the output feedback control, but it is much smaller than the nonadaptive case. We state that the impedance control based on output feedback can realize the system objective and track the desired trajectory successfully, where the system error is converging to a small value close to zero.

Based on the above simulation cases, it is obvious that the adaptive NN impedance control is effective in handling the unknown nonlinear dynamic system with saturation effect. When the end effector is required to move in free space, the robotic manipulator has a good tracking performance. When the robotic manipulator is in contact with the obstacle, the adaptive NN impedance control has a better performance than PD control evidently. Meanwhile, the state feedback control

has a slightly better tracking performance than the output feedback scheme. Nevertheless, some output information may not be measurable in the state feedback scheme, which means the output feedback scheme is more practical than the state feedback scheme.

V. CONCLUSION

In this paper, impedance control based on NNs has been developed to solve the interaction problem for a robotic manipulator. Both full state and output feedback controllers have been considered with an auxiliary signal to handle the saturation effect of a robotic manipulator. Stability of the closed loop system has been proved by Lyapunov's direct method. Simulation results have been provided to show that the proposed control is able to track a desired impedance model with a good performance.

ACKNOWLEDGMENT

The authors would like to thank the Editor-in-Chief, the Associate Editor, and the anonymous reviewers for their constructive comments which helped to improve the quality and presentation of this paper.

APPENDIX

Proof for Theorem 1: Multiplying (27) by $e^{\kappa t}$ yields

$$\frac{d}{dt}(V_2 e^{\kappa t}) \leq C e^{\kappa t}.$$

Integrating the above inequality, we obtain

$$V_n \leq \left(V_2(0) - \frac{C}{\kappa} e^{-\kappa t} + \frac{C}{\kappa} \right) \leq V_2(0) + \frac{C}{\kappa}.$$

For z_1 , we have

$$\frac{1}{2} z_1^T z_1 \leq V_2(0) + \frac{C}{\kappa}.$$

Then, we can obtain

$$\|z_1\| \leq \sqrt{B}$$

where $B = 2(V_2^*(0) + C/\kappa)$. ■

REFERENCES

- [1] S. S. Ge, T. H. Lee, and C. J. Harris, *Adaptive Neural Network Control of Robotic Manipulators*. London, U.K.: World Sci., 1998.
- [2] N. Hogan, "Impedance control: An approach to manipulation. Part I: Theory; Part II: Implementation; Part III: Applications," *J. Dyn. Syst. Meas. Control*, vol. 107, no. 1, pp. 8–16, 1985.
- [3] C. C. de Wit, G. Bastin, and B. Siciliano, *Theory of Robot Control*. London, U.K.: Springer, 1996.
- [4] N. Hogan, "On the stability of manipulators performing contact tasks," *IEEE J. Robot. Autom.*, vol. 4, no. 6, pp. 677–686, Dec. 1988.
- [5] W.-Z. Gao and R. R. Selmic, "Neural network control of a class of nonlinear systems with actuator saturation," *IEEE Trans. Neural Netw.*, vol. 17, no. 1, pp. 147–156, Jan. 2006.
- [6] C. Yang *et al.*, "Human-like adaptation of force and impedance in stable and unstable interactions," *IEEE Trans. Robot.*, vol. 27, no. 5, pp. 918–930, Oct. 2011.
- [7] E. Burdet, R. Osu, D. W. Franklin, T. E. Milner, and M. Kawato, "The central nervous system stabilizes unstable dynamics by learning optimal impedance," *Nature*, vol. 414, no. 6862, pp. 446–449, 2001.
- [8] H. Kazerooni, P. K. Houpt, and T. B. Sheridan, "Robust compliant motion for manipulators. Part I: The fundamental concepts of compliant motion; Part II: Design method," *IEEE J. Robot. Autom.*, vol. 2, no. 2, pp. 83–105, Jun. 1986.
- [9] C.-C. Cheah and D.-W. Wang, "Learning impedance control for robotic manipulators," *IEEE Trans. Robot. Autom.*, vol. 14, no. 3, pp. 452–465, Jun. 1998.
- [10] W.-S. Lu and Q.-H. Meng, "Impedance control with adaptation for robotic manipulations," *IEEE Trans. Robot. Autom.*, vol. 7, no. 3, pp. 408–415, Jun. 1991.
- [11] Y. H. Kim and F. L. Lewis, "Reinforcement adaptive learning neural-net-based friction compensation control for high speed and precision," *IEEE Trans. Control Syst. Technol.*, vol. 8, no. 1, pp. 118–126, Jan. 2000.
- [12] B. Xu, "Robust adaptive neural control of flexible hypersonic flight vehicle with dead-zone input nonlinearity," *Nonlin. Dyn.*, vol. 80, no. 3, pp. 1509–1520, 2015.
- [13] B. Xu, Z. Shi, C. Yang, and F. Sun, "Composite neural dynamic surface control of a class of uncertain nonlinear systems in strict-feedback form," *IEEE Trans. Cybern.*, vol. 44, no. 12, pp. 2626–2634, Dec. 2014.
- [14] L. Liu, D. Liang, and C. Liu, "Nonlinear state-observer control for projective synchronization of a fractional-order hyperchaotic system," *Nonlin. Dyn.*, vol. 69, no. 4, pp. 1929–1939, 2012.
- [15] W. He, S. S. Ge, Y. Li, E. Chew, and Y. S. Ng, "Neural network control of a rehabilitation robot by state and output feedback," *J. Intell. Robot. Syst.*, vol. 80, no. 1, pp. 15–31, Oct. 2015.
- [16] R. Colbaugh, H. Seraji, and K. Glass, "Direct adaptive impedance control of robot manipulators," *J. Robot. Syst.*, vol. 10, no. 2, pp. 217–248, 1993.
- [17] C. Yang, Z. Li, and J. Li, "Trajectory planning and optimized adaptive control for a class of wheeled inverted pendulum vehicle models," *IEEE Trans. Cybern.*, vol. 43, no. 1, pp. 24–36, Feb. 2013.
- [18] S. S. Ge, C. C. Hang, L.-C. Woon, and X.-Q. Chen, "Impedance control of robot manipulators using adaptive neural networks," *Int. J. Intell. Control Syst.*, vol. 2, no. 3, pp. 433–452, 1998.
- [19] Z. Li, S. S. Ge, and A. Ming, "Adaptive robust motion/force control of holonomic-constrained nonholonomic mobile manipulators," *IEEE Trans. Syst., Man, Cybern. B, Cybern.*, vol. 37, no. 3, pp. 607–616, Jun. 2007.
- [20] Z. Liu, F. Wang, Y. Zhang, X. Chen, and C. L. P. Chen, "Adaptive fuzzy output-feedback controller design for nonlinear systems via backstepping and small-gain approach," *IEEE Trans. Cybern.*, vol. 44, no. 10, pp. 1714–1725, Oct. 2014.
- [21] Y.-J. Liu, L. Tang, S. Tong, and C. L. P. Chen, "Adaptive NN controller design for a class of nonlinear MIMO discrete-time systems," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 26, no. 5, pp. 1007–1018, May 2015.
- [22] Z. Li, S. S. Ge, and S. Liu, "Contact-force distribution optimization and control for quadruped robots using both gradient and adaptive neural networks," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 25, no. 8, pp. 1460–1473, Aug. 2014.
- [23] S.-L. Dai, C. Wang, and M. Wang, "Dynamic learning from adaptive neural network control of a class of nonaffine nonlinear systems," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 25, no. 1, pp. 111–123, Jan. 2014.
- [24] C. L. P. Chen, Y.-J. Liu, and G.-X. Wen, "Fuzzy neural network-based adaptive control for a class of uncertain nonlinear stochastic systems," *IEEE Trans. Cybern.*, vol. 44, no. 5, pp. 583–593, May 2014.
- [25] F. L. Lewis, S. Jagannathan, and A. Yeldirek, *Neural Network Control of Robot Manipulators and Nonlinear Systems*. London, U.K.: Taylor and Francis, 1999.
- [26] M. Chen, S. S. Ge, and B. Ren, "Robust attitude control of helicopters with actuator dynamics using neural networks," *IET Control Theory Appl.*, vol. 4, no. 12, pp. 2837–2854, Dec. 2010.
- [27] M. Gaeta, V. Loia, F. Orciuoli, and P. Ritrovato, "S-WOLF: Semantic workplace learning framework," *IEEE Trans. Syst., Man, Cybern., Syst.*, vol. 45, no. 1, pp. 56–72, Jan. 2015.
- [28] Z. Li and C.-Y. Su, "Neural-adaptive control of single-master-multiple-slaves teleoperation for coordinated multiple mobile manipulators with time-varying communication delays and input uncertainties," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 24, no. 9, pp. 1400–1413, Sep. 2013.
- [29] S.-L. Dai, C. Wang, and F. Luo, "Identification and learning control of ocean surface ship using neural networks," *IEEE Trans. Ind. Informat.*, vol. 8, no. 4, pp. 801–810, Nov. 2012.
- [30] Z. Liu, C. Chen, Y. Zhang, and C. L. P. Chen, "Adaptive neural control for dual-arm coordination of humanoid robot with unknown nonlinearities in output mechanism," *IEEE Trans. Cybern.*, vol. 45, no. 3, pp. 521–532, Mar. 2015.
- [31] L. Wang *et al.*, "Energy-efficient SVM learning control system for biped walking robots," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 24, no. 5, pp. 831–837, May 2013.
- [32] S.-C. Tong, Y.-M. Li, and H.-G. Zhang, "Adaptive neural network decentralized backstepping output-feedback control for nonlinear large-scale systems with time delays," *IEEE Trans. Neural Netw.*, vol. 22, no. 7, pp. 1073–1086, Jul. 2011.
- [33] Y.-J. Liu, S.-C. Tong, D. Wang, T.-S. Li, and C. L. P. Chen, "Adaptive neural output feedback controller design with reduced-order observer for a class of uncertain nonlinear SISO systems," *IEEE Trans. Neural Netw.*, vol. 22, no. 8, pp. 1328–1334, Aug. 2011.
- [34] Y.-J. Liu, C. L. P. Chen, G.-X. Wen, and S. C. Tong, "Adaptive neural output feedback tracking control for a class of uncertain discrete-time nonlinear systems," *IEEE Trans. Neural Netw.*, vol. 22, no. 7, pp. 1162–1167, Jul. 2011.
- [35] R. Cui, B. Ren, and S. S. Ge, "Synchronised tracking control of multi-agent system with high order dynamics," *IET Control Theory Appl.*, vol. 6, no. 5, pp. 603–614, Mar. 2012.
- [36] R. Cui, J. Guo, and Z. Mao, "Adaptive backstepping control of wheeled inverted pendulums models," *Nonlin. Dyn.*, vol. 79, no. 1, pp. 501–511, 2015.
- [37] Y.-J. Liu, S. Tong, and C. L. P. Chen, "Adaptive fuzzy control via observer design for uncertain nonlinear systems with unmodeled dynamics," *IEEE Trans. Fuzzy Syst.*, vol. 21, no. 2, pp. 275–288, Apr. 2013.
- [38] V. Santibañez, K. Camarillo, J. Moreno-Valenzuela, and R. Campa, "A practical PID regulator with bounded torques for robot manipulators," *Int. J. Control Autom. Syst.*, vol. 8, no. 3, pp. 544–555, 2010.
- [39] J. Huang, C. Wen, W. Wang, and Z.-P. Jiang, "Adaptive stabilization and tracking control of a nonholonomic mobile robot with input saturation and disturbance," *Syst. Control Lett.*, vol. 62, no. 3, pp. 234–241, 2013.
- [40] J.-X. Xu, Y. Tan, and T.-H. Lee, "Iterative learning control design based on composite energy function with input saturation," *Automatica*, vol. 40, no. 8, pp. 1371–1377, 2004.

- [41] M. Chen, B. Jiang, J. Zou, and X. Feng, "Robust adaptive tracking control of the underwater robot with input nonlinearity using neural networks," *Int. J. Comput. Intell. Syst.*, vol. 3, no. 5, pp. 646–655, 2010.
- [42] Y.-J. Liu and S.-C. Tong, "Adaptive NN tracking control of uncertain nonlinear discrete-time systems with nonaffine dead-zone input," *IEEE Trans. Cybern.*, vol. 45, no. 3, pp. 497–505, Mar. 2015.
- [43] Z. Liu, G. Lai, Y. Zhang, X. Chen, and C. L. P. Chen, "Adaptive neural control for a class of nonlinear time-varying delay systems with unknown hysteresis," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 25, no. 12, pp. 2129–2140, Dec. 2014.
- [44] W. He, Y. Chen, and Z. Yin, "Adaptive neural network control of an uncertain robot with full-state constraints," *IEEE Trans. Cybern.*, to be published.
- [45] M. Chen, S. S. Ge, and B. Ren, "Adaptive tracking control of uncertain MIMO nonlinear systems with input constraints," *Automatica*, vol. 47, no. 3, pp. 452–465, 2011.
- [46] S.-C. Tong and Y.-M. Li, "Adaptive fuzzy output feedback tracking backstepping control of strict-feedback nonlinear systems with unknown dead zones," *IEEE Trans. Fuzzy Syst.*, vol. 20, no. 1, pp. 168–180, Feb. 2012.
- [47] Z. Guo, J. Wang, and Z. Yan, "Global exponential synchronization of two memristor-based recurrent neural networks with time delays via static or dynamic coupling," *IEEE Trans. Syst., Man, Cybern., Syst.*, vol. 45, no. 2, pp. 235–249, Feb. 2015.
- [48] C. L. P. Chen, G.-X. Wen, Y.-J. Liu, and F.-Y. Wang, "Adaptive consensus control for a class of nonlinear multiagent time-delay systems using neural networks," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 25, no. 6, pp. 1217–1226, Jun. 2014.
- [49] C. Wen, J. Zhou, Z.-T. Liu, and H.-Y. Su, "Robust adaptive control of uncertain nonlinear systems in the presence of input saturation and external disturbance," *IEEE Trans. Autom. Control*, vol. 56, no. 7, pp. 1672–1678, Jul. 2011.
- [50] W. He and S. S. Ge, "Vibration control of a flexible beam with output constraint," *IEEE Trans. Ind. Electron.*, vol. 62, no. 8, pp. 5023–5030, Aug. 2015.
- [51] W. He, S. Zhang, and S. S. Ge, "Adaptive control of a flexible crane system with the boundary output constraint," *IEEE Trans. Ind. Electron.*, vol. 61, no. 8, pp. 4126–4133, Aug. 2014.
- [52] W. He and S. S. Ge, "Vibration control of a flexible string with both boundary input and output constraints," *IEEE Trans. Control Syst. Technol.*, vol. 23, no. 4, pp. 1245–1254, Jul. 2015.
- [53] W. He, C. Sun, and S. S. Ge, "Top tension control of a flexible marine riser by using integral-barrier Lyapunov function," *IEEE/ASME Trans. Mechatronics*, vol. 20, no. 2, pp. 497–505, Apr. 2015.
- [54] S. S. Ge and C. Wang, "Adaptive neural network control of uncertain MIMO non-linear systems," *IEEE Trans. Neural Netw.*, vol. 15, no. 3, pp. 674–692, May 2004.
- [55] S. S. Ge, C. C. Hang, T. H. Lee, and T. Zhang, *Stable Adaptive Neural Network Control*. Boston, MA, USA: Kluwer Academic, 2001.
- [56] R. Horn and C. Johnson, *Matrix Analysis*. Cambridge, U.K.: Cambridge Univ. Press, 1990.
- [57] W. He, S. S. Ge, B. V. E. How, Y. S. Choo, and K.-S. Hong, "Robust adaptive boundary control of a flexible marine riser with vessel dynamics," *Automatica*, vol. 47, no. 4, pp. 722–732, 2011.
- [58] W. He, S. Zhang, and S. S. Ge, "Robust adaptive control of a thruster assisted position mooring system," *Automatica*, vol. 50, no. 7, pp. 1843–1851, 2014.
- [59] Y. Li, S. S. Ge, and C. Yang, "Learning impedance control for physical robot–environment interaction," *Int. J. Control*, vol. 85, no. 2, pp. 182–193, 2012.
- [60] S. Behtash, "Robust output tracking for nonlinear system," *Int. J. Control*, vol. 51, no. 6, pp. 931–933, 1990.



Wei He (S'09–M'12) received the B.Eng. degree from the College of Automation Science and Engineering, South China University of Technology, Guangzhou, China, in 2006, and the Ph.D. degree from the National University of Singapore (NUS), Singapore, in 2011.

He was a Research Fellow with the Department of Electrical and Computer Engineering, NUS, from 2011 to 2012. He is currently with the Center for Robotics and the School of Automation Engineering, University of Electronic Science and Technology of China, Chengdu, China. His current research interests include robotics, distributed parameter systems, and intelligent control systems.

Dr. He has served as an Editor for the *Journal of Intelligent and Robotic Systems* (Springer).



Yiting Dong (S'14) received the B.Eng. degree from the School of Automation Engineering, University of Electronic Science and Technology of China, Chengdu, China, in 2014, where he is currently pursuing the M.Eng. degree.

His current research interests include robotics, neural network control, and adaptive control.



Changyin Sun received the M.S. and Ph.D. degrees in electrical engineering from Southeast University, Nanjing, China, in 2001 and 2003, respectively.

He is a Distinguished Professor with the School of Automation, Southeast University and the University of Science and Technology Beijing, Beijing, China. His current research interests include intelligent control, flight control, pattern recognition, and optimal theory.